

OUJDA ANGADS, Morocco

WMO: 601150

Lat: 34.787N

Lon: 1.924W

Elev: 468

StdP: 95.83

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.0	2.3	-4.9	2.6	11.5	-2.9	3.1	11.5	15.1	15.7	13.4	14.4	1.8	240	0.482

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.4	38.1	20.6	36.1	20.7	34.2	20.5	23.9	32.4	23.1	31.2	22.4	29.8	5.8	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.7	17.3	26.7	20.9	16.5	26.4	20.1	15.7	25.7	74.4	32.5	71.0	31.2	68.0	29.7	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.5	9.9	8.9	-1.1	42.7	1.4	1.9	-2.1	44.1	-2.9	45.2	-3.7	46.3	-4.7	47.7		
			-1.9	25.8	1.5	1.1	-3.0	26.5	-3.9	27.2	-4.7	27.8	-5.8	28.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.1	10.7	11.3	13.5	15.6	19.1	23.2	26.6	27.0	23.2	19.9	14.6	11.7
	DBStd	6.37	2.81	2.82	2.79	2.78	3.13	2.87	2.67	2.44	2.60	2.97	3.25	2.84
	HDD10.0	64	26	17	4	0	0	0	0	0	0	0	2	15
	HDD18.3	1052	238	197	151	92	28	1	0	0	1	17	121	208
	CDD10.0	3007	46	53	113	167	281	397	514	526	396	307	139	66
	CDD18.3	954	0	1	2	9	51	147	256	268	147	66	8	1
	CDH23.3	10152	2	7	61	157	589	1580	2947	2932	1233	571	64	9
	CDH26.7	4922	0	2	14	39	212	730	1634	1593	501	184	12	1
Wind	WSAvg	3.7	3.9	3.7	3.6	3.7	3.8	3.8	3.9	3.8	3.4	3.2	3.6	3.6
Precipitation	PrecAvg	335	29	36	44	45	30	12	2	4	16	28	33	39
	PrecMax	540	104	130	139	182	164	77	13	28	90	154	118	181
	PrecMin	109	0	0	2	0	0	0	0	0	0	0	0	0
	PrecStd	104	25	32	36	40	36	18	3	7	20	35	26	46
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.9	24.0	28.7	30.9	35.0	38.1	41.4	41.2	36.8	34.1	28.7	24.0
		MCWB	12.1	13.1	15.3	16.8	17.8	20.2	20.9	20.4	20.6	18.0	15.6	13.2
	2%	DB	20.2	21.1	24.9	27.3	31.1	35.1	38.2	38.1	33.2	30.9	25.0	21.0
		MCWB	11.7	12.3	14.1	16.2	17.7	20.6	20.8	20.7	20.0	17.3	14.4	12.6
	5%	DB	18.2	19.2	22.5	24.9	28.9	33.0	36.1	35.9	31.2	28.7	22.8	19.0
		MCWB	11.3	12.1	13.8	15.3	17.5	20.0	20.7	21.1	19.8	17.7	14.3	12.2
	10%	DB	16.6	17.4	20.2	22.7	26.7	30.9	34.1	34.0	29.7	26.7	20.9	17.2
		MCWB	10.9	11.6	13.5	14.7	16.9	19.4	20.6	21.0	19.6	17.5	13.9	11.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.2	15.0	17.2	19.2	21.0	23.7	24.7	25.2	23.6	21.2	18.4	15.3
		MCDB	18.8	20.4	23.8	26.7	29.7	33.1	34.2	33.8	30.3	28.0	22.7	19.5
	2%	WB	13.1	13.9	15.8	17.5	19.6	22.4	23.7	24.1	22.3	20.0	16.8	14.0
		MCDB	17.9	18.7	22.0	24.2	27.4	31.3	33.3	31.9	28.6	25.8	21.7	18.6
	5%	WB	12.2	13.0	14.8	16.3	18.5	21.4	22.9	23.4	21.5	19.1	15.7	13.0
		MCDB	16.9	17.9	20.6	22.7	26.1	29.8	31.6	30.8	27.5	25.2	20.5	17.6
	10%	WB	11.2	12.0	13.9	15.2	17.5	20.2	22.0	22.6	20.8	18.3	14.6	12.1
		MCDB	15.7	16.5	19.0	20.9	24.6	28.2	30.0	29.6	26.5	24.1	19.2	16.4
Mean Daily Temperature Range	5% DB	MDBR	11.2	11.7	12.6	13.2	13.5	14.2	14.7	14.4	13.0	12.8	11.4	11.2
		MCDBR	14.0	14.9	17.2	17.6	17.7	17.2	17.7	17.0	15.5	15.9	14.1	13.5
		MCWBR	7.9	8.4	9.0	8.8	7.8	7.2	7.0	6.8	6.6	6.8	6.8	7.3
	5% WB	MCDBR	12.0	13.2	14.6	15.0	14.8	15.2	14.9	14.1	12.8	13.4	12.1	11.8
		MCWBR	7.4	8.1	8.5	8.1	7.1	6.9	6.6	6.1	6.0	6.4	6.4	7.1
Clear-Sky Solar Irradiance	taub	0.299	0.326	0.383	0.409	0.444	0.476	0.526	0.499	0.451	0.395	0.343	0.310	
	taud	2.525	2.450	2.278	2.211	2.136	2.044	1.911	1.994	2.134	2.295	2.442	2.518	
	Ebn at Noon	914	920	887	877	847	817	772	786	806	827	849	877	
	Edh at Noon	81	98	126	142	155	170	193	174	144	113	88	78	
All-Sky Solar Radiation	RadAvg	2.90	3.80	5.03	6.10	6.96	7.70	7.58	6.84	5.56	4.24	3.07	2.65	
	RadStd	0.26	0.31	0.33	0.43	0.48	0.24	0.18	0.21	0.25	0.27	0.25	0.19	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.48	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	+0.22	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page